

FINDINGS DOCUMENT

Inventory Transaction Visibility — SQL Analysis Results

ClearView Medical Supplies | Week 1 Mini Project

Document Version	v1.0
Date Completed	2026/0519
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Stakeholder	Naledi Mokoena, Head of Procurement & Inventory
Dataset	Inventory_Transactions.csv — 1,415 rows, 9 columns
Tool Used	MySQL Workbench + Power BI

1. Data Overview

Before running any queries, the following was confirmed about the dataset:

Attribute	Detail
Total rows	1,415 transactions
Date range	January 2025 to September 2025
Locations	North, South, East, West
SKU categories	SURG, LENS, FRM, SOL
Risk levels	Low Risk (418), Medium Risk (805), High Risk (192)
Lead time values	7, 14, 30, 45 days (categorical)
Staff members	Optician_A, Optician_B, Optician_C, Surgeon_K

2. Data Cleaning Summary

The following issues were identified and resolved before importing into MySQL:

#	Issue Found	Fix Applied	Status
1	Semicolon delimiter (not comma)	Data was fixed when using Power Query	Resolved
2	Revenue column stored as text	Changed data type to Decimal Number via Power Query using English (US) locale	Resolved

3. Query Findings

Each use case from the URD was tested using MySQL. Results, screenshots, and plain-language interpretations are recorded below.

UC-01: All High Risk Transactions

Query used:

```
SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level
FROM inventory_transactions WHERE risk_level = 'High Risk' ORDER BY lead_time
DESC;
```

Query result screenshot:

The screenshot displays the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'sakila' database with tables 'inventory_transactions' and 'surgical_optical_supplies'. The 'Columns' pane for 'inventory_transactions' lists: order_id (text), date (text), SKU (text), quantity_sold (int), location (text), staff_id (text), revenue (double), lead_time (int), and risk_level (text). The 'Query' pane shows the following SQL query:

```
4. Identify high revenue orders with long lead times
/*
View all high risk transactions*/
SELECT
order_id,
date,
SKU,
quantity_sold,
location,
lead_time,
risk_level
FROM inventory_transactions
WHERE risk_level= 'High Risk'
ORDER BY lead_time DESC;
```

The 'Result Grid' pane shows the following data:

order_id	date	SKU	quantity_sold	location	lead_time	risk_level
ORD-507	8/12/2025	SURG-4-01	10	North	45	High Risk
ORD-515	3/4/2025	SURG-4-01	1	East	45	High Risk
ORD-517	10/4/2025	SURG-4-01	23	West	45	High Risk
ORD-520	4/28/2025	SURG-4-01	6	South	45	High Risk
ORD-523	11/13/2025	SURG-4-01	2	East	45	High Risk
ORD-525	1/3/2025	SURG-4-01	25	West	45	High Risk
ORD-531	10/2/2025	SURG-4-01	8	West	45	High Risk
ORD-533	8/23/2025	SURG-4-01	18	South	45	High Risk
ORD-545	2/31/2025	SURG-4-01	8	North	45	High Risk
Inventory_transactions 17						

The 'Output' pane shows the following messages:

```
74 00:36:32 SELECT * FROM inventory_transactions WHERE date NOT REGEXP "[0-9]{1,2}/[0-9]{1,2}/[0-9]{4}" LIMIT 1000
75 00:58:49 SELECT * FROM inventory_transactions WHERE date NOT REGEXP "[0-9]{1,2}/[0-9]{1,2}/[0-9]{4}" LIMIT 1000
76 01:09:17 SELECT * FROM inventory_transactions WHERE risk_level= 'High Risk' AND lead_time<= 45 LIMIT 0.1000
77 22:06:13 --1. View all high risk transactions SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level F...
78 22:06:44 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WH...
79 22:08:14 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WH...
```

Result summary:

Field	Value
Total rows returned	192
Key column(s) observed	SKU
Unexpected findings	N/A

Finding: 192 transactions were flagged as high risk, they all had a lead time of 45 days, SURG was the only SKU that came up.

Recommendation: Seeing all high risk products are SURG, with a lead time 45 days. These products should be prioritised and moving forward be ordered 6 weeks in advance.

UC-02: Top 10 Largest Orders by Quantity

Query used:

```
SELECT order_id, SKU, quantity_sold, location, staff_id, revenue FROM inventory_transactions ORDER BY quantity_sold DESC LIMIT 10;
```

Query result screenshot:

The screenshot displays the MySQL Workbench interface. The 'Query' tab is active, showing a SQL query that filters for 'High Risk' items and then selects the top 10 largest orders by quantity. The 'Result Grid' shows the results of the query, with columns: order_id, SKU, quantity_sold, location, staff_id, and revenue. The results are sorted by quantity_sold in descending order. The first row shows order_id 1669, SKU LENS-D-01, quantity_sold 25, location South, staff_id Optician_C, and revenue 312.5. The second row shows order_id 769, SKU SURG-P-02, quantity_sold 25, location South, staff_id Optician_C, and revenue 2125. The third row shows order_id 603, SKU SURG-P-02, quantity_sold 25, location East, staff_id Optician_C, and revenue 2125. The fourth row shows order_id 1001, SKU LENS-M-02, quantity_sold 25, location West, staff_id Optician_B, and revenue 625. The fifth row shows order_id 1876, SKU FRM-M-10, quantity_sold 25, location East, staff_id Optician_C, and revenue 1125. The sixth row shows order_id 918, SKU LENS-D-01, quantity_sold 25, location East, staff_id Optician_B, and revenue 312.5. The seventh row shows order_id 1633, SKU SURG-P-02, quantity_sold 25, location East, staff_id Optician_A, and revenue 2125. The eighth row shows order_id 763, SKU FRM-M-10, quantity_sold 25, location West, staff_id Surgeon_K, and revenue 1125. The ninth row shows order_id 1223, SKU FRM-M-10, quantity_sold 25, location South, staff_id Optician_B, and revenue 1125. The tenth row shows order_id 1669, SKU LENS-D-01, quantity_sold 25, location South, staff_id Optician_C, and revenue 312.5. The 'Output' tab shows the execution of the query, with a message indicating that the query was successful and returned 192 rows.

Result summary:

Field	Value
Total rows returned	10
Key column(s) observed	SKU, Quantity sold, Revenue
Unexpected findings	The North branch does not come up in top 10 largest order by quantity

Finding: The 10 largest order were a range of products namely FRM,SURG,LENS. Most of these orders were placed by an Optician.

Recommendation: These three products move fast so it is best we keep close eye on stock levels and order frequently to meet demand.

UC-03: Medium and High Risk Orders at South Branch

Query used:

```
SELECT order_id, date, SKU, quantity_sold, lead_time, risk_level FROM
inventory_transactions WHERE location = 'South' AND (risk_level = 'Medium
Risk' OR risk_level = 'High Risk') ORDER BY risk_level, lead_time DESC;
```

Query result screenshot:

The screenshot displays the MySQL Workbench interface. The SQL editor contains the following query:

```
/*3.Filter transactions by locations and risk level*/
SELECT
order_id,
date,
SKU,
quantity_sold,
location,
lead_time,
risk_level
FROM inventory_transactions
WHERE location='South'
AND (risk_level='Medium Risk' OR risk_level='High Risk')
ORDER BY risk_level,lead_time DESC;
```

The Results Grid shows the following data:

order_id	date	SKU	quantity_sold	location	lead_time	risk_level
ORD-853	2/2/2025	SURG-4-01	23	South	45	High Risk
ORD-1607	3/28/2025	SURG-4-01	2	South	45	High Risk
ORD-520	4/28/2025	SURG-4-01	6	South	45	High Risk
ORD-1909	4/14/2025	SURG-4-01	17	South	45	High Risk
ORD-533	8/23/2025	SURG-4-01	18	South	45	High Risk
ORD-1605	4/5/2025	SURG-4-01	4	South	45	High Risk
ORD-547	12/5/2025	SURG-4-01	18	South	45	High Risk
ORD-698	10/27/2025	SURG-4-01	4	South	45	High Risk
ORD-1235	11/25/2025	SURG-4-01	14	South	45	High Risk
ORD-1601	4/14/2024	SURG-4-01	17	South	45	High Risk

The Output tab shows the following messages:

```
96 22:37:16 SELECT order_id, SKU, quantity_sold, location, staff_id, revenue FROM inventory_transactions ORDER BY a... 10 row(s) returned 0.015 sec / 0.000 sec
97 23:04:33 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WHE... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec
98 23:04:56 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WHE... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec
99 23:06:28 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WHE... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec
100 23:07:09 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WHE... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se... 0.000 sec
101 23:08:30 SELECT order_id, date, SKU, quantity_sold, location, lead_time, risk_level FROM inventory_transactions WHE... 272 row(s) returned 0.016 sec / 0.000 sec
```

Result summary:

Field	Value
Total rows returned	272
Key column(s) observed	Risk level, location
Unexpected findings	N/A

Finding: The 3 products that came up in this query was SURG,LENS,FRM. The south branch had 209 Medium risk and 63 High risk transactions.

Recommendation: The orders the South branch manager should prioritise are(in this order) SURG,FRM,LENS. SURG being a high risk item should be looked at closely and ordered in advance.

UC-04: High Revenue Orders with Long Lead Times

Query used:

```
SELECT order_id, date, SKU, location, revenue, lead_time, risk_level FROM
inventory_transactions WHERE revenue > 500 AND lead_time >= 30 ORDER BY
revenue DESC;
```

Query result screenshot:

The screenshot displays the MySQL Workbench interface. The 'Query' tab is active, showing a SQL query that filters for orders with revenue greater than 500 and lead time greater than or equal to 30, ordered by revenue in descending order. The 'Result Grid' shows the first 25 rows of the query results. The columns are: order_id, date, SKU, location, revenue, lead_time, and risk_level. The results show various orders from different locations (West, South, East) with varying revenue and lead times, all categorized as 'High Risk'.

order_id	date	SKU	location	revenue	lead_time	risk_level
ORD-525	1/7/2025	SURG-4-01	West	3750	45	High Risk
ORD-571	4/22/2025	SURG-4-01	South	3750	45	High Risk
ORD-647	10/8/2025	SURG-4-01	South	3750	45	High Risk
ORD-915	3/31/2025	SURG-4-01	West	3750	45	High Risk
ORD-1061	12/9/2025	SURG-4-01	East	3750	45	High Risk
ORD-1082	11/15/2025	SURG-4-01	East	3750	45	High Risk
ORD-1120	5/10/2025	SURG-4-01	West	3750	45	High Risk
ORD-1458	5/20/2025	SURG-4-01	North	3750	45	High Risk
ORD-1912	4/24/2025	SURG-4-01	East	3750	45	High Risk
ORD-1912	7/17/2024	SURG-4-01	South	3750	45	High Risk

The 'Output' tab at the bottom shows the execution of the query, indicating that 272 rows were returned. The 'Message' pane shows several error messages related to the query execution, including 'Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'WHERE...' at line 1'.

Result summary:

Field	Value
Total rows returned	575
Key column(s) observed	Revenue, lead times
Unexpected findings	N/A

Finding: This query gave us the highest quantity sold, corresponding lead times and highest revenue sold for a single order which was R3750. The item was a high risk SKU and it was SURG product.

Recommendation: My recommendation would be to order SURG products in advance as they have the highest lead time and products with a lead time of 30 days should have minimum safety stock and when that safety stock reaches below the minimum it should trigger a reorder for the 30 lead time products.

4. Overall Summary of Findings

The following key insights were surfaced across all four use cases:

- 192 High Risk orders all carry a 45-day lead time, making them the most urgent replenishment priority
SURG SKUs dominate both the High Risk and high-revenue categories
- The South branch has the highest transaction volume and the most risk exposure
- Clear view needs to add a new column, stock levels the floor and safety stock

5. Recommendations for Naledi

Based on the analysis, the following actions are recommended:

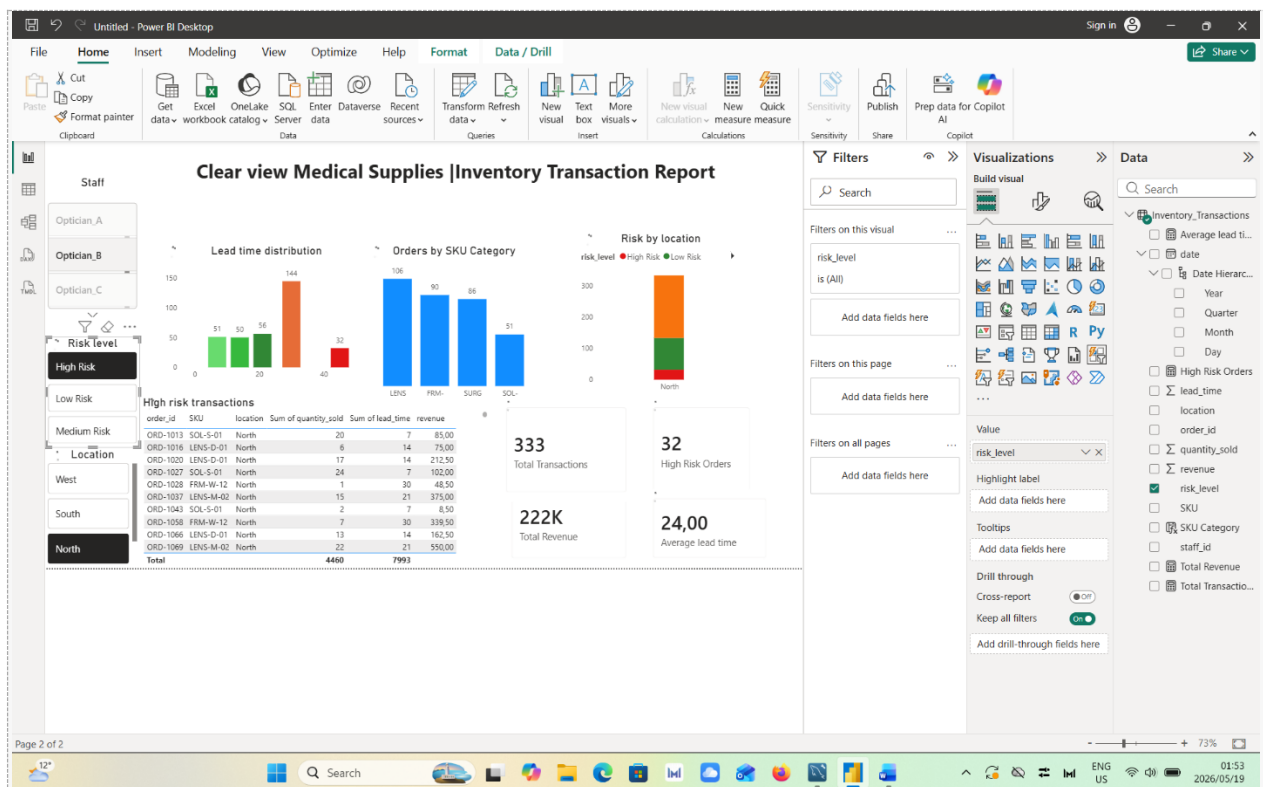
#	Recommendation	Owner	Priority
1	Trigger reorder for all High Risk SURG SKUs immediately	Procurement Team	High
2	Set a weekly review cadence for the High Risk transaction list	Branch Managers	High
3	Flag any order with revenue > R500 and lead time >= 30 days for escalation]	Naledi Mokoena	Medium
4	Add 2 more columns to keep 30 day lead time item stock levels healthy	Procurement Team	Medium

6. Power BI Dashboard

The findings above were visualised in a Power BI report containing the following elements:

- 4 KPI cards: Total Transactions, Total Revenue, Average Lead Time, High Risk Orders
- Bar chart: Orders by Risk Level (Low / Medium / High)
- Bar chart: Revenue by SKU Category (SURG / LENS / FRM / SOL)
- Stacked bar chart: Risk Level by Location
- Column chart: Lead Time Distribution (7 / 14 / 30 / 45 days)
- Table visual: High Risk Transactions filtered list
- Slicers: Location, Risk Level, Staff ID

Dashboard screenshot:



7. Sign Off

	Analyst	Stakeholder (Naledi)
Name	Kgomotso Mokhorro	Naledi Mokoena

Date	2026/05/19	2026/05/19
Signature		